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APPLICATION ARTICLE

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# Advancing plant metabolic research by using large language models to expand databases and extract labeled data

**Rachel Knapp**[1] | **Braidon Johnson**[1,2] | **Lucas Busta**[1]

[1]Department of Chemistry and Biochemistry, University of Minnesota Duluth, Duluth, Minnesota, USA

[2]Department of Chemical Engineering, University of Minnesota Duluth, Duluth, Minnesota, USA

**Correspondence**

Lucas Busta, Department of Chemistry and Biochemistry, University of Minnesota Duluth, Duluth, Minnesota, USA.
Email: bust0037@d.umn.edu

## Abstract

**Premise:** Recently, plant science has seen transformative advances in scalable data collection for sequence and chemical data. These large datasets, combined with machine learning, have demonstrated that conducting plant metabolic research on large scales yields remarkable insights. A key next step in increasing scale has been revealed with the advent of accessible large language models, which, even in their early stages, can distill structured data from the literature. This brings us closer to creating specialized databases that consolidate virtually all published knowledge on a topic.

**Methods:** Here, we first test different combinations of prompt engineering techniques and language models in the identification of validated enzyme–product pairs. Next, we evaluate the application of automated prompt engineering and retrieval-augmented generation to identify compound–species associations. Finally, we build and determine the accuracy of a multimodal language model–based pipeline that transcribes images of tables into machine-readable formats.

**Results:** When tuned for each specific task, these methods perform with high (80–90%) or modest (50%) accuracies for enzyme–product pair identification and table image transcription, but with lower false-negative rates than previous methods (decreasing from 55% to 40%) for compound–species pair identification.

**Discussion:** We enumerate several suggestions for researchers working with language models, among which is the importance of the user's domain-specific expertise and knowledge.

KEYWORDS

artificial intelligence, plant metabolism, structured data extraction

Public databases are powerful tools that researchers can use to enhance studies of plant chemical diversity and metabolism. Significant efforts have been made to develop and populate resources such as Phytozome, GenBank, and UniProt, which support research into plant genomes and proteomes (Goodstein et al., 2012; Benson et al., 2013; UniProt Consortium, 2015); the Plant Metabolic Network and the Kyoto Encyclopedia of Genes and Genomes (KEGG), which support metabolic research (Kanehisa et al., 2012; Hawkins et al., 2021); Global Natural Products Social Molecular Networking (GNPS), MassBank, and MassBank of North America, which support mass spectrometry research (Horai et al., 2010; Wang et al., 2016); and Natural Products Online's LOTUS, which aids in studying the distribution of plant

Rachel Knapp and Braidon Johnson contributed equally to this work.

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natural products across phylogenies (Rutz et al., 2022). These databases are essential for foundational tasks in plant metabolic research, such as identifying biosynthetic enzymes and tracking their evolution, identifying compounds in plant extracts, and understanding how plant products are distributed across the plant kingdom. Despite their utility, however, these databases remain incomplete. Significant amounts of data remain locked in PDFs of scientific articles or in supplementary files, and current databases may not meet users' specific needs for data structure or curation, particularly in lineages that lack well-studied model or crop plants. This contributes to the challenges faced by researchers when assembling, refining, and curating the information necessary for project-specific analyses. Addressing these limitations is crucial, otherwise researchers will continue to face ongoing obstacles in leveraging existing data, likely hindering advances in plant biochemistry, natural product discovery, and metabolic engineering.

To overcome challenges related to data mining and database structures, researchers from diverse fields are increasingly using large language models (LLMs). These models serve a variety of purposes: embedding models transform sequences (such as words or proteins) into numerical vectors for analysis; chat/completion models generate coherent, context-aware text; classification models label input data based on predefined categories; and multimodal models integrate multiple data types, like text and images, for tasks such as image captioning. Some models can also follow user-defined prompts—text-based instructions that provide detail on how the language model should perform a given task. Many recent studies report on prompt engineering strategies to improve the accuracy of a model's output. Examples of these strategies include few-shot and multi-shot learning, where the model is given one or more examples of question–answer pairs, and techniques like chain-of-thought and Tree of Thoughts, which guide the model through step-by-step reasoning for complex tasks (Wei et al., 2022; Yao et al., 2023). Inspired by these approaches, researchers in the natural sciences, including the field of plant chemistry, are already using language models to tackle data mining and database-related problems. For instance, some are summarizing large literature landscapes (González-Márquez et al., 2024), while others are extracting information about enzyme–substrate pairs (Smith et al., 2024), linking specific phytochemicals to specific plant species through data mining (Busta et al., 2024), and identifying yet unreported enzyme activities from relatively understudied domains of life (Samusevich et al., 2024). These examples illustrate the transformative potential of language models and their ability to advance our field, but also highlight that we are at the very beginning when it comes to integrating language models into plant metabolic research.

Here, we provide several examples of how language models can help plant scientists address challenges related to extracting and organizing data from unstructured or loosely structured sources. We also report results from experiments with language models of different levels of sophistication (including models that can process images) as well as with various prompt engineering strategies. First, we used prompt engineering to extract enzyme–product relationships from National Center for Biotechnology Information (NCBI) entries, enabling the separation of validated enzyme–product pairs from predicted enzyme–product pairs. Next, we developed an automated prompt engineering strategy to improve the extraction of chemical compound–plant species relationships from text. Finally, we integrated vision models with prompts and language model calls to extract structured data from PNG images of tables in research article PDF files. Our approaches achieved high levels of accuracy and illustrate the potential of language models in data organization projects. The present work also highlights the critical importance of model users' detailed knowledge and subject matter expertise in plant metabolism.

## METHODS

Enzyme and compound information used as data in the first stage of our study (see below under "Extracting validated enzyme–product relationships") was retrieved from NCBI's PubChem and Protein databases via the NCBI API and processed in R using the rentrez package (Winter, 2017). For data handling and visualization, tidyverse (Wickham et al., 2019) and ggplot2 (Wickham, 2016) were used together with ggupset (Ahlmann-Eltze, 2024) for plotting. All significance testing was conducted using the statix package (Kassambara, 2024). The completion language models used in this study were OpenAI models accessed via the OpenAI API (OpenAI, San Francisco, California, USA). In this initial stage, we manually developed system prompts for these models based on iterative testing of the prompts' performance. For the next stage (see below under "Automated prompt engineering for compound–species relationship detection"), we used GPT-4 to generate prompts, using WandB (Biewald, 2020) to automate their creation and test prompt performance across a manually curated set of previously published test cases (Busta et al., 2024). Embeddings were created with the BAAI/bge-small-en-v1.5 embedding model downloaded from HuggingFace (https://huggingface.co/BAAI/bge-small-en-v1.5). Retrieval of passages for the retrieval-augmented generation (RAG) system was performed by having GPT-4 generate a few sentences resembling an academic answer to the question, converting those into embeddings, and then using a nearest-neighbor algorithm (sklearn.neighbors.NearestNeighbors) to find the most similar matches in the vector database of abstracts. For the final stage (see below under "Vision AI for quantitative data table transcription"), several Python modules were used to help transcribe and process table images by turning them into structured JSON formats including OS for file handling, base64 for image encoding, requests for API interactions, pandas for data manipulation, and OpenAI for model integration. The PNG images of tables were base64-encoded and analyzed using OpenAI's GPT-4o model. That same model

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was used to check if both plant species and compound names were present in the initial transcriptions and to determine the table's orientation—whether rows represented species or compounds—to ensure accurate data extraction. The R package ggtree (Yu et al., 2017) was used to visualize the phylogenetic tree. All code and raw data used in this project are available at https://github.com/thebustalab/ai_in_phytochemistry (see Data Availability Statement).

## RESULTS AND DISCUSSION

### Extracting validated enzyme–product relationships

The goal in the initial stage of our study was to assess the ability of LLMs to distinguish between database records that report the validated product of an enzyme catalyst (positive entries) versus those that report a predicted product (negative entries). The first step towards this goal was to obtain a collection of database records with both positive and negative representatives. To obtain this collection, we first conducted command-line NCBI searches using three enzyme-related search terms ("beta-amyrin synthase", "lupeol synthase", "cycloartenol synthase") and added the first 20 records from each search into our collection, resulting in an initial set of 60 records identified by their protein IDs (Figure 1A). Manual inspection of these 60 records revealed an imbalance, with a predominance of negative records (reports of enzymes and their predicted, not validated, products), giving an initial ratio of positive to negative records of roughly 1:4. To create a more balanced collection, we supplemented the initial set of 60 with additional records that we identified based on manual inspection of peer-reviewed articles that reported the validated products of specific enzymes. After these steps, our collection consisted of 142 records, including 93 positive records and 49 negative records, and a final ratio of positive to negative records of roughly 2:1.

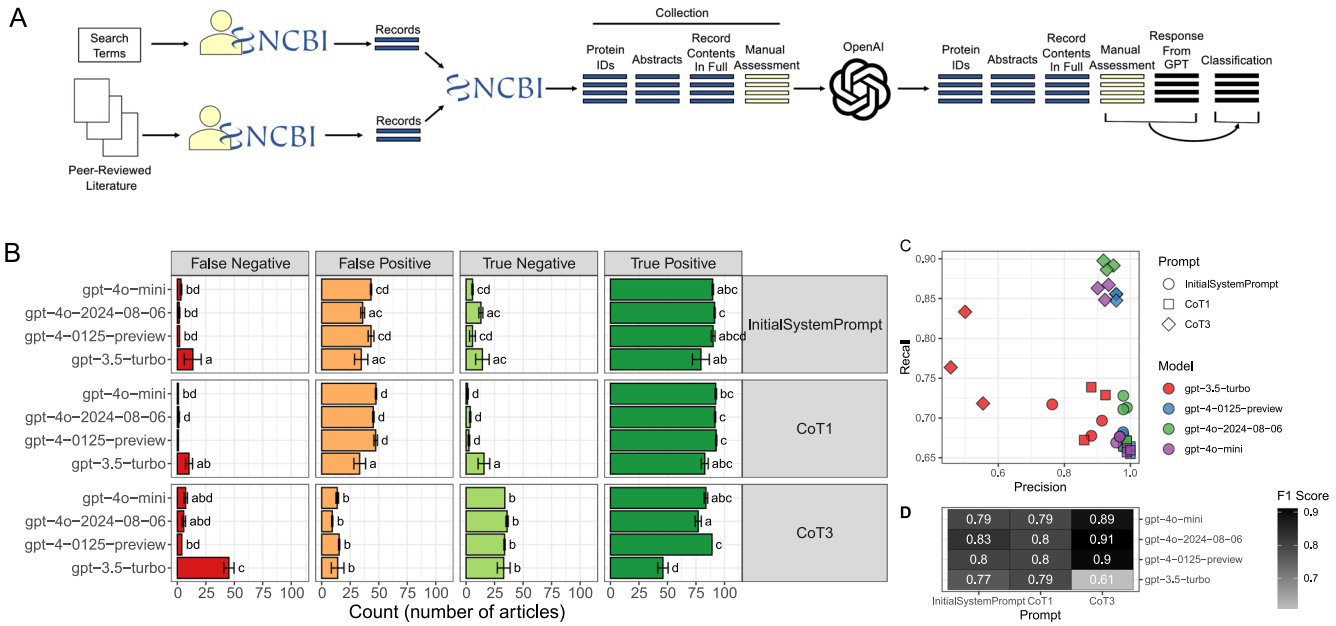


**FIGURE 1** Workflow and results of protein–product relationship evaluation. (A) Workflow used to generate results. The search terms indicate the enzymes searched for in NCBI (e.g., "beta-amyrin synthase" or "cycloartenol synthase"). Peer-reviewed literature indicates the manually curated NCBI Pubchem and Protein databases listing verified protein–product pairs. The collection is made up of both the products of the search terms and the contents of the curated database. Blue lines indicate NCBI protein records, yellow lines indicate cells created through manual evaluation, and black boxes represent cells generated via language model response and comparison between language model response and manual evaluation. (B) The outcomes of language model classification of protein–product pairs. The *y*-axis represents the different models used, and the *x*-axis represents the number of entries evaluated. Red indicates false negatives, orange denotes false positives, light green signifies true negatives, and dark green indicates true positives. In this case, a true positive indicates a record the language model correctly labeled as a positive, a true negative indicates a record the model correctly labeled as a negative, a false positive is a record the model erroneously labeled as a positive, and a false negative is a record that the model erroneously labeled as a negative. Bar heights and error bars indicate the mean and standard deviation of $n = 3$ replicates, respectively. In some cases, a model–prompt pair generated the same outcome in all three replicates, leading to standard deviation values of zero and thus appearing in this figure as a bar without an error bar. Lowercase letters shown to the right of each bar indicate the group to which each belongs based on a Tukey honestly significant difference (HSD) test. Tests compared the number of articles identified as false negative, false positive, true negative, or true positive across all models. Thus, letters should be compared across all vertical sets of panels; bars that do not share a letter in common are significantly different from one another. (C) A plot showing the precision and recall scores for the various prompt and model combinations. Shape indicates prompt, and color indicates model as shown in the legend. There are multiple entries for each model–prompt combination to show the precision and recall of each independent ("replicate") analysis where $n = 3$. (D) A heat map showing the mean F1 scores for each prompt and model combination. The F1 score is the harmonic mean of precision and recall, that is, 2(precision × recall)/(precision + recall). CoT, chain-of-thought.

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With a collection of manually validated positive and negative records in hand, we next retrieved the full contents of each record, including the PubMed IDs of any articles describing the enzyme in the record. In a separate step, we retrieved the abstracts of the articles with those PubMed IDs and associated those abstracts with the corresponding records in our collection. Thus, at this stage, our collection consisted of the 142 records, each with their Protein ID, full record contents, associated abstracts, and manual annotation as a positive or negative record (Figure 1A). Finally, we looped over all the records in our collection and passed them one by one to an OpenAI language model alongside a system prompt designed to elicit a response from the model labeling the record as positive or negative. We compared the label assigned by the language model against our manually assigned label, and, considering the manually assigned label to be accurate, labeled each language model response as a true positive, true negative, false positive, or false negative. In this case, a true positive indicates a record the language model correctly labeled as a positive, a true negative indicates a record the model correctly labeled as a negative, a false positive is a record the model erroneously labeled as a positive, and a false negative a record that the model erroneously labeled as a negative.

Using the approach above, we tested the accuracy of several language models under the instruction of several different styles of system prompts. A system prompt is a model input that describes a task that the language model should perform. System prompts can range from very simple to very complex and have a substantial impact on the efficacy of the model (Schick and Schütze, 2021; Grabb, 2023; Zhang et al., 2024). Furthermore, through the process of prompt engineering, a prompt can be optimized to increase the system's performance against a given metric (Diab et al., 2022). A key example emphasizing the importance of prompt engineering is the "chain-of-thought" prompting technique developed in 2022 (Wei et al., 2022). The most basic form of the method consists of simply adding the words "Let's think through this step-by-step" into an existing prompt. The researchers found that by adding these words to a prompt, model performance increased substantially (from 18% to 80% accuracy on the MultiArith test using OpenAI GPT-3) (Wei et al., 2022). The chain-of-thought prompting method has since become a standard prompting technique for increasing language model performance, to the extent that OpenAI has incorporated it as a default behavior into some of its most recent models (Wang et al., 2024). Here, we evaluated the accuracy of four different language models (GPT-3.5 Turbo, GPT-4o version 2024-08-06, GPT-4-0125 Preview, and GPT-4o mini) using one of three different system prompts (a standard, non-chain-of-thought prompt; a "let's think step-by-step" chain-of-thought prompt [CoT1]; and a more detailed chain-of-thought prompt [CoT3] that asked the model to include intermediate steps in its output). Of importance here are the relative sizes (number of parameters) of these models, with GPT-3.5 Turbo having many fewer parameters (around 175 billion parameters) and GPT-4 models having potentially an order of magnitude more (exact numbers are an OpenAI trade secret and are not available). Details of all system prompts used in this section are provided in Appendix S1, but all prompts were geared toward asking the model "Does the enzyme described in the record make the product?"

In the tests above, the best-performing prompt/model combination (as determined by the percentage of correct answers, i.e., accuracy, which is defined as true positives + true negatives/total tests) was GPT-4o version 2024-08-06 with CoT3, which had an accuracy of 89.5%. The worst-performing combination was GPT-3.5 Turbo with CoT3, which had an accuracy of 54.6%. It should be noted that because the ratio of positive to negative records in the reference benchmark set is roughly 2:1, it is expected that accurate models will classify more records as positive than negative. After completing the tests described above, we also noticed a number of interesting patterns in the results, several of which pertained to differences in performance between models. We observed that GPT-3.5 Turbo often generated significantly more false negatives than the other models when responding to the same prompt (Figure 1B). We also found that, generally speaking, the GPT-4 models were largely comparable (Figure 1B). This suggests that model size, which is generally thought of as being correlated with model sophistication, is connected to the ability of the model to distinguish between true positive results and false negative results. Others have also observed improvement in project-specific tasks when using larger versus smaller models (Park et al., 2023; Yang et al., 2024; Zhong et al., 2024; Herck et al., 2025). Up-to-date information about model performance on benchmark tasks can be found at, for example: https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard.

Next, we noticed several patterns in the results connected to the prompts that were used. Above all, we observed that when using the most sophisticated chain-of-thought prompt, the models were able to achieve a significantly higher rate of true negative detection, which seemed to be connected to a lower false positive rate (Figure 1B). In other words, when using a sophisticated system prompt, all the models tested here were more frequently able to recognize instances in which an entry contained an enzyme that had been experimentally validated to make the associated product, rather than just predicted to make the product.

To verify that the trends observed above were not likely to be impacted if the number of records in the reference benchmark set were to be increased, we repeated the classification steps above using randomly selected subsets of the 142 records present in the reference set. Three randomly selected subsets of 100 records each were analyzed using the language model and prompt combinations in the same way, and only those statistical differences that were present in the tests of all three of the subsets and the test of the full set of 142 records are included in Figure 1B. This suggests that the conclusions drawn above are

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unlikely to be impacted if additional records were included in our benchmark set.

Next, we examined precision and recall scores for each model and prompt combination. Precision is the proportion of true positive cases among all the cases the model recognizes as positives, while recall is the proportion of true cases that are recognized as positives by the model. We noted that the typical tradeoff between precision and recall was present in the tests we conducted, that is, for most models and prompts, precision increased as recall decreased. However, there were some outliers in the case of the detailed chain-of-thought prompt. Under this prompting regime, all of the GPT-4 models, but not the GPT-3.5 model, exhibited both high precision (>0.9) and high recall (>0.85; Figure 1C). We also examined the F1 scores, the harmonic mean of precision and recall, to get a general idea about the performance of each prompt and model combination in the present context. This metric also revealed that the more detailed chain-of-thought prompt in combination with one of the GPT-4 models led to the highest performance (F1 score >0.89; Figure 1D) while, interestingly, that same prompt with the GPT-3.5 model led to the poorest performance (F1 score = 0.61).

Considered together, our results suggest several major points. First, a general chain-of-thought prompt like "let's think step by step" performs significantly worse than an explicit chain-of-thought prompt detailing all the steps that a model should take in generating its answer. In particular, it seems that this more detailed form of a chain-of-thought prompt leads the model to recognize false positives and correctly classify them as true negatives. This more detailed chain-of-thought prompt, in combination with newer, larger models, led to the highest performance, approaching 90% accuracy in distinguishing between validated versus predicted enzyme–product associations. This contrasts with the combination of the smaller model (GPT-3.5 Turbo) with a detailed chain-of-thought prompt, which performed only slightly better than chance, with 54% accuracy. It seems that model + prompt combinations will need to be evaluated on a case-by-case basis before deployment in a scientific research project.

## Automated prompt engineering for compound–species relationship detection

Recognizing the critical role of prompt engineering in our analysis of extracting protein–product relationships from NCBI, our next objective was to explore how systematic prompt engineering and related strategies could enhance task-specific accuracy. In a previous study, we used a collection of >2500 manually evaluated articles to test the ability of LLMs to identify text passages describing the identification of specific chemical compounds in extracts of particular plant species (Busta et al., 2024). This is a challenging task because accurate performance requires recognizing instances in which (i) the name of a chemical compound co-occurs in the same body of text with the name of a plant species, but the text provides no evidence for the plant species actually making that chemical compound in the text, and distinguishing such instances from (ii) instances in which the chemical and species names occur in a context that does indicate evidence for the chemical's production by the plant species. The best-performing language model in that study successfully processed these text passages with a low false positive rate—indicating that it rarely misattributed a specific chemical to a plant species. However, this came at the expense of a high false negative rate, meaning the model often missed true instances of chemical production by particular species. Therefore, our goal in this stage of the study was to determine if prompt engineering and related strategies could improve this high false negative rate.

For systematic prompt engineering, we implemented a method that leveraged language models to generate a diverse array of prompts. This method involved two language models: the first, termed the prompt generator language model, was responsible for generating prompts, while the second, the prompt evaluator language model, assessed the generated prompts against cases with established answers. Initially, the prompt generator language model created 50 prompts, which we then tested for accuracy using the evaluator language model and 20 test cases. The accuracy of these prompts ranged from 65% to 90%, with the majority achieving an accuracy of either 75% or 80% (Figure 2A, inset histogram).

To investigate the reasons behind the observed differences in prompt performance, we employed an embedding model to generate numeric vectors representing the semantic content of each prompt. We then applied principal component analysis to these vectors, allowing us to map the prompts within a two-dimensional space defined by the first two principal components (Figure 2A), and used *k*-means clustering to identify clusters within this principal component space. Silhouette scores, which measure how similar each point is to others within its cluster compared to those in other clusters, indicated that a seven-cluster model provided a reasonable fit for the data distribution in the principal component space (Figure 2B). Interestingly, this linguistic-based separation in principal component space also correlated with performance outcomes: high-performing prompts predominantly clustered on the left side, while lower-performing prompts clustered on the right. In particular, the leftmost cluster (Cluster 4) and the rightmost cluster (Cluster 3) demonstrated significantly different performance levels (Wilcoxon test, $P < 0.05$). These results suggest that linguistic variations in prompt phrasing can lead to substantial performance differences, even among prompts designed for the same task, a phenomenon that has been reported by others (Zhou et al., 2023; Smith et al., 2024). Together, our results and those in the literature strongly suggest that a diverse range of task-specific prompts should be tested in language model applications before deploying a large-scale pipeline.

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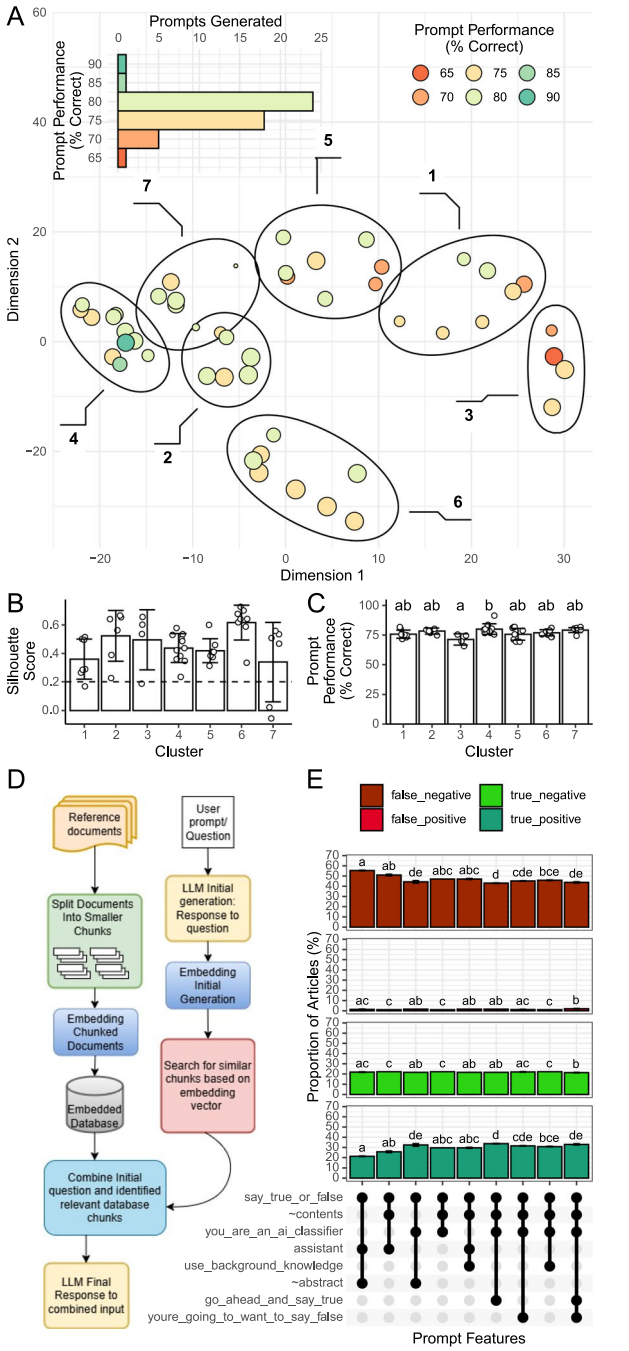


**FIGURE 2** Automated prompt engineering and retrieval-augmented generation (RAG) to enhance the identification of compound–species associations. (A) Performance (inset histogram) and principal component analysis of the linguistic meaning of 50 prompts generated for the task of compound–species relationship identification. Ellipses indicate clusters of linguistically similar prompts. (B) Silhouette scores for each point in each cluster, where scores >0.5 suggest strong cluster membership, scores between 0.2 and 0.5 indicate moderate support, and scores below 0.2 show low cluster association. (C) Performance of the prompts in each cluster. Bar heights and error bars indicate means and standard deviations of $n = 3$ replicates, respectively. Letters indicate statistical groups determined by Tukey's HSD test (adjusted $P < 0.05$). (D) Model of the RAG system. (E) Performance of different system prompts in combination with RAG. Contents of the system prompts are indicated on the $x$-axis. Colors indicate true/false positives/negatives, as described for Figure 1.

Alongside the prompt engineering experiments described above, we also evaluated a well-known language model technique called retrieval-augmented generation (RAG) (Lewis et al., 2021; Wu et al., 2024), which enhances language model outputs by supplying domain-specific information alongside the input prompt. A basic implementation of RAG involves creating a system where a user poses a question and the system retrieves relevant excerpts from a content database to provide as additional context to the language model for response generation (Figure 2D). In our study, we used the manually evaluated abstracts from a prior study as the content database ("Reference documents," Figure 2D; Busta et al., 2024), from which the RAG system retrieved information to use alongside each question as input to the language model. For example, we asked questions like "Is compound X produced by species Y?" and included the abstract excerpts retrieved by the RAG system as part of a structured prompt to the language model to assess the question. This approach not only provides the language model with clear instructions through a detailed system prompt, but also supports response generation with peer-reviewed information from the content database. The full prompts used in this section are provided in Appendix S2.

Using the RAG system described above, along with a set of modular system prompts, we evaluated the performance of the language model GPT-4o. For each combination of content (abstract or retrieved content) and system prompt, we evaluated the model's performance in 200 compound–species associations for which we had previously manually determined the answers. We then compared the model's responses against the manually determined correct answers and calculated true positive, false positive, true negative, and false negative rates for each prompt and content combination. The largest effect we observed was when the system prompt provided a detailed description of the role that the model should play. For example, instructing the model that it was simply a "helpful assistant" yielded results similar to what we had observed previously, with a 55% false negative rate (Busta et al., 2024). However, when we told the model to act as an "AI classifier" and provided further details about how it should perform that role, we saw a significant reduction in the false negative rate (down to ~40% from ~55%). Others have reported on the importance of role- and goal-oriented prompts in creating effective language model pipelines (Li et al., 2024).

Next, we evaluated the classifications of the language model–based approach used here against classifications obtained based simply upon the co-occurrence of a chemical name and a plant species name in the same text. For the co-occurrence classifier, we wrote an R script that classified abstracts in which the relevant chemical name and plant species name co-occurred as positive associations and abstracts that lacked co-occurrence as negative associations (script available at https://github.com/thebustalab/ai_in_phytochemistry). The co-occurrence script was case-insensitive. The best-performing language model classification system described above ("Say True

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or False" + "AI classifier" + "Abstract") performed considerably better than co-occurrence, with lower false negative and false positive rates (43% vs. 54% and 2% vs. 7%, respectively; Appendix S3) and higher true negative and true positive rates (22% vs. 16% and 34% vs. 22%, respectively).

In the effort being described here, we did detect some smaller effects due to differences in providing the abstract versus retrieved content or adding additional modular components in the system prompt that attempted to encourage the model to describe compound–species relationships as true (an attempt to reduce the false negative rate). However, these effects were minor compared to the impact of providing a well-defined role as part of the system prompt. Thus, taken together with the literature, our experiments suggest that while providing additional, perhaps redundant, content to the model does not substantially improve the false negative rate, prompt engineering and prompt specifics have a substantial and significant impact. We suggest that future tests include styles of system prompt engineering that involve few-shot and multi-shot learning. These are prompting styles in which examples of how to handle common and edge cases are provided to the model, increasing its ability to accurately handle task-specific activities. Finally, another avenue that may be worth exploration is the implementation of small language models in processes like that described here. Small language models have considerably smaller computing requirements than LLMs and can often be run on consumer-grade hardware. Emerging evidence suggests that for specific tasks, as opposed to general purposes, small language models can compete with LLMs in terms of their performance (Guo et al., 2023; Luo et al., 2023; Bolton et al., 2024).

## Vision AI for quantitative data table transcription

Our final objective was to assess the potential of multimodal language models to generate structured data relevant to plant metabolic research. Substantial amounts of quantitative data on phytochemical profiles are stored in tables within scientific articles. In some instances, these tables are provided in a machine-readable format, such as CSV files in supplementary materials. However, such files are not always provided, particularly not by older publications. Consequently, tables embedded in PDF files represent a valuable yet underutilized source of information that is labor-intensive to systematically compile into a larger structured dataset for analysis. Without developing automated or semi-automated methods to unify this information, we risk losing the results gathered over decades of research.

To evaluate the ability of multimodal language models to transcribe quantitative data from tables published in scientific articles, we began by collecting 13 complex tables from a set of 13 different scientific articles describing the abundances of phytosterols in the seed oils of various plant species (Stuchlik and Zak, 2002; Ryan et al., 2007; Kamal-Eldin and Moazzami, 2009; Ngila et al., 2009; Hassanein et al., 2014; Kozlowska et al., 2016; Islam et al., 2017; García-González et al., 2018; Bettaieb et al., 2019; Yang et al., 2019; Nguyen et al., 2020; Silva et al., 2020; Saini et al., 2021). The tables in these articles reported the abundances of phytosterols in 67 different plant species, with many species being described by multiple tables in the set. Initially, we attempted to use the multimodal model to extract data from PNG images of these table images directly, but we found that reproducible and accurate transcription was severely hindered by the non-uniform structure of the tables. In their raw form, many of the tables included units of scientific measurements in row names, reports of ranges rather than means and standard deviations, and complex layouts with summary statistics or multiple presentations of the same data in different units, including percentages. The poor performance of the model in this environment of non-uniformity led us to explore strategies to improve accuracy.

To improve table transcription by multimodal language models, we first manually simplified the images by importing the tables into image editing software and removing portions that appeared to confuse the model. These included table captions and rows or columns containing summary statistics. We also replaced the common names of plant species in each table with their Latin names as reported in the respective articles, then saved each table as a high-resolution PNG image. Next, we divided the transcription process into two phases (Figure 3A). The first phase was image extraction, where PNG images of each table were encoded as base64 images and then passed to a multimodal language model (OpenAI GPT-4o) that transcribed the table as a CSV file. The resulting CSV data was then passed to a second language model to verify the presence of chemical names or plant species names. If no recognizable names were present, the multimodal model was prompted to attempt the transcription again. Otherwise, the CSV file containing both compound and species names was written to disk and, if necessary, automatically transposed so that compound names were organized as row headers and species names as column headers.

In the second phase of our pipeline, each CSV file was processed using a combination of language model calls and Python modules (Figure 3A). First, each CSV file was read from the disk and passed to a language model that re-transcribed the table in a simpler format that omitted symbols, standard deviations, and other non-standard characters. The output from that language model was then checked by another language model, which verified the presence of chemical compound names and species names before writing the table to a CSV file and conducting transposition if necessary. This sequence of steps resulted in processed CSV tables, which were then manually compared against the original images. This manual comparison revealed that the processed tables were 98% accurate compared to the original images (Figure 3B). Overall, our pipeline and its results suggest that, although some manual intervention is required, multimodal models are highly

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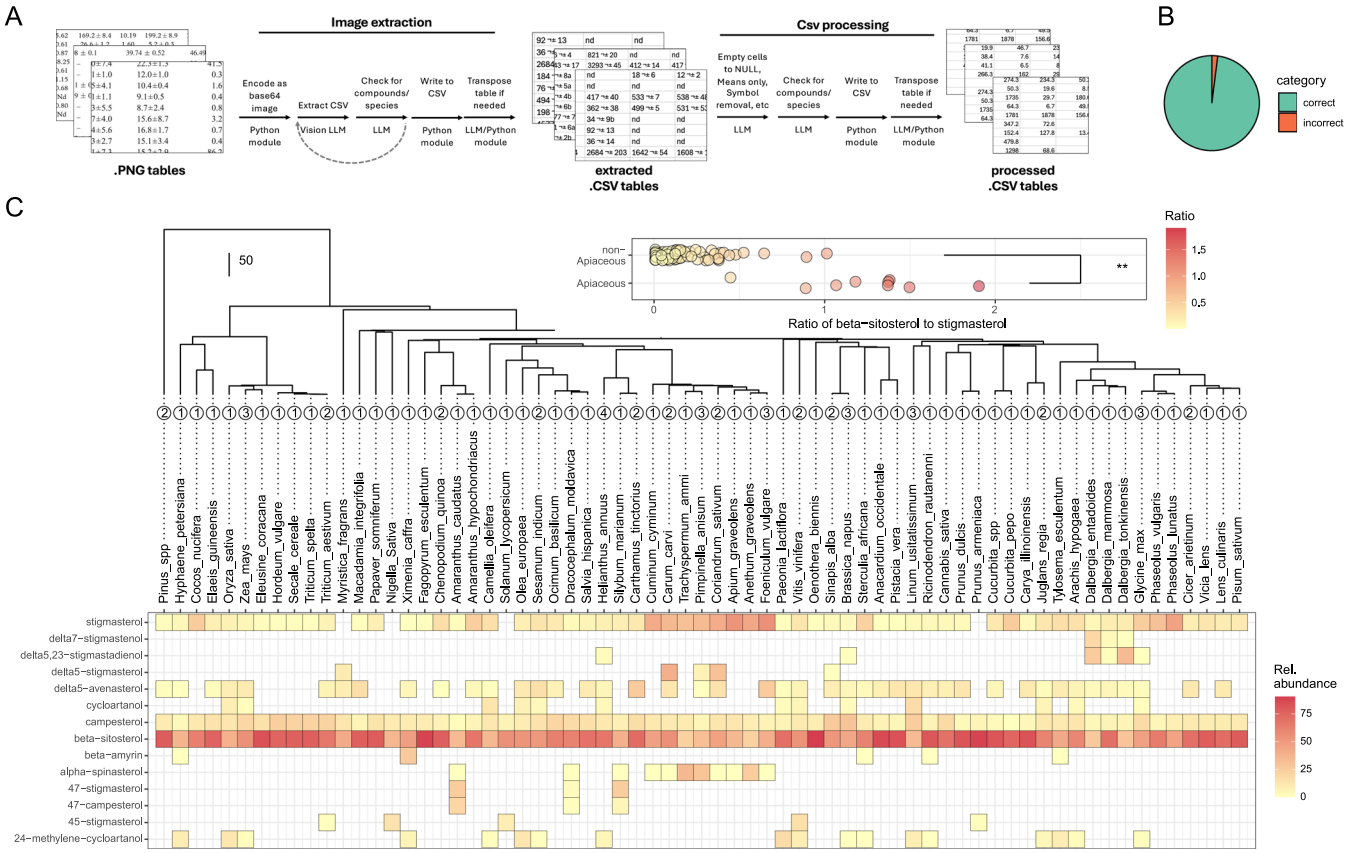


**FIGURE 3** Workflow and results from using a multimodal language model to transcribe images of scientific tables. (A) Workflow used to transcribe/extract and process text from images of tables. (B) Pie chart indicating the accuracy of the transcription process based on manual comparison of transcribed data and the original tables. (C) Phylochemical map showing the abundance (in the heat map, abundance is indicated in the legend) of different phytosterols across the species included in this study (shown in the phylogenetic tree). The phylogeny was prepared by pruning a previously published megaphylogeny (Qian and Jin, 2015). The inset plot shows the ratio of beta-sitosterol to stigmasterol in apiaceous versus non-apiaceous species. Stars indicate statistically significant differences in the mean ratio between the two groups (Wilcoxon test, $P < 0.05$).

useful for creating structured data from previously reported research results.

Next, we compared the output of the multimodal language model pipeline described above against the output of a leading open-source optical character recognition tool, PyTesseract (https://pypi.org/project/pytesseract/). We passed PyTesseract PNG images that had been processed manually as described above (i.e., replacement of common names with Latin names, removal of rows or columns that contained summary statistics, etc.). Overall, the optical character recognition tool struggled in a variety of ways, including with recognizing column headers that spanned multiple lines, letters present in cells to represent statistical test outcomes, and special characters present in row or header names (Appendix S4). In contrast, the language model handled such cases well.

To illustrate one possible application of (semi-) automated table transcription by LLMs, we next placed the transcribed quantitative data into a phylogenetic context. We pruned a megaphylogeny (Qian and Jin, 2015) to create a phylogeny for the 67 species reported across the transcribed tables. We then created a heat map next to the tree showing the relative abundance of different phytosterols in the seed oils of those species, based on the table data (Figure 3C). This revealed that, in most species, beta-sitosterol was the most abundant phytosterol; however, in some species, stigmasterol was also a prominent component. We examined the ratio of stigmasterol to beta-sitosterol in each species for which both compounds had been reported. Interestingly, transcribed data from three different publications consistently showed that species in the carrot family (Apiaceae) had significantly higher ratios of stigmasterol to beta-sitosterol, suggesting that a lineage-specific event may have occurred in the ancestor of this group, leading to altered phytosterol ratios in seed oils (Figure 3C, inset).

While developing the multimodal language model–based transcription pipeline, we made several observations that will likely impact future implementations of similar systems. Perhaps our most important observation was the considerable variation in table structure between publications. Tables varied greatly and often included forms of normalized or summarized data interspersed with abundance measurements. A primary challenge in developing this pipeline was determining

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how to systematically distinguish between abundance values and other types of information in table cells. In the end, we overcame this difficulty through manual annotation. Having to manually inspect and polish the tables is perhaps beneficial because it forces human intervention and makes the researcher notice details like which solvents were used, whether different plant varieties were tested, or if various growing conditions were applied. Being aware of this sort of data for the species or chemicals of interest seems advantageous, albeit at the cost of throughput.

We also noticed that using high-resolution images greatly increased the accuracy of language model transcription. Low-resolution images were almost always transcribed incorrectly, with inaccuracies in both numerical transcription and the assignment of particular numbers to specific table cell positions. Interestingly, we found that table size did not seem to impact accuracy; a 3 × 5 table was processed with equal accuracy as a 10 × 14 table, provided the images of the tables are of sufficient resolution. This suggests that, as long as high-resolution images can be obtained, this workflow should be suitable for large table sizes. However, scaling this pipeline to handle a large number of tables will require automated or semi-automated solutions for managing the non-uniform data types presented in quantitative tables.

## CONCLUSIONS

Here, we conducted several explorations and evaluations of LLMs in applications related to creating structured data from unstructured or largely unstructured information sources. Our key findings indicated that prompt content at both a general architectural level (standard prompts vs. chain-of-thought prompts) and at a word level significantly affect the performance of a language model. Carefully and specifically defining the role that a language model should play in the context of performing a particular task also has an impact on the model's accuracy. When using a multimodal model, we noticed that high-resolution images led to higher transcription accuracy. However, complex table structures—including summary statistics and other information within a given table—led to difficulties in transcribing and merging multiple tables into a single unified format. These difficulties, and our approach to overcoming them, highlight the importance of human involvement and subject matter expertise in pipelines that involve language models.

Overall, our results illustrate that LLMs have considerable promise but are not without limitations. Our observations demonstrate that prompts need to be designed in a task-specific manner for language models to perform with high accuracy and appropriately handle edge cases, and future implementations should be designed accordingly. We also note that designing such prompts is a skill that will likely be useful in many disciplines and could serve as a valuable learning objective in relevant coursework (Schulhoff et al., 2024). Similarly, collaboration between disciplines—including plant sciences, linguistics, data science, and bioinformatics—could lead to advances in this area and facilitate the integration of language models with existing databases and tools that are important in plant metabolic research.

Looking forward, we note the growing trend of integrating language models with other related entities and approaches, including machine learning and knowledge graphs (Lim et al., 2023). It is also interesting to note the reported efficacy of small (as opposed to large) language models in highly discipline-specific tasks (Guo et al., 2023; Luo et al., 2023; Bolton et al., 2024). The development of multimodal models may also influence our field in the future as models capable of interpreting both text and biological sequence data are developed (Xu et al., 2023; Chen et al., 2024). In all these cases, the importance of labeled data is clear, and we would like to highlight the need to support efforts geared toward generating such data. Creating such datasets can not only help us in using existing language models most effectively in our discipline but could also pave the way for the creation of discipline-specific models that perform at high levels and support advances in plant metabolic research.

### AUTHOR CONTRIBUTIONS

L.B. conceived the research. All authors designed and conducted experiments. All authors acquired and analyzed the data and prepared the figures. L.B. led the writing of the manuscript, and all authors made contributions. All authors approved the final version of the manuscript.

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to hold the University of Minnesota Duluth accountable to American Indian peoples and nations.

## DATA AVAILABILITY STATEMENT

All data and code used in this study are available for free download from the project repository (https://github.com/thebustalab/ai_in_phytochemistry). The prompts used in identifying enzyme–product and compound–species pairs are provided in the Supporting Information.

## ORCID

*Lucas Busta* http://orcid.org/0000-0002-0102-9986

## REFERENCES

Ahlmann-Eltze, C. 2024. ggupset: Combination matrix axis for "ggplot2" to create "UpSet" plots. R package version 0.4.0.9000. Available at https://github.com/const-ae/ggupset [accessed 11 April 2025].

Benson, D. A., M. Cavanaugh, K. Clark, I. Karsch-Mizrachi, D. J. Lipman, J. Ostell, and E. W. Sayers. 2013. GenBank. *Nucleic Acids Research* 41: D36–D42.

Bettaieb, I., S. Bourgou, P. Detry, W. Wannes, T. Kenny, R. Ksouri, I. Hamrouni, and M. L. Fauconnier. 2019. Green extraction of fennel and anise edible oils using bio-based solvent and supercritical fluid: Assessment of chemical composition, antioxidant property, and oxidative stability. *Food and Bioprocess Technology* 12: 1798–1807.

Biewald, L. 2020. Experiment tracking with Weights and Biases. Available at https://www.wandb.com/ [accessed 11 April 2025].

Bolton, E., A. Venigalla, M. Yasunaga, D. Hall, B. Xiong, T. Lee, R. Daneshjou, et al. 2024. BiomedLM: A 2.7B parameter language model trained on biomedical text. arXiv:2403.18421. Available at https://doi.org/10.48550/arXiv.2403.18421 [posted 27 March 2024; accessed 11 April 2025].

Busta, L., D. Hall, B. Johnson, M. Schaut, C. Hanson, A. Gupta, M. Gundrum, et al. 2024. Mapping of specialized metabolite terms onto a plant phylogeny using text mining and large language models. *The Plant Journal* 120: 406–419.

Chen, Z., T. Chen, C. Xie, Y. Xue, X. Zhang, J. Zhou, and X. Fang. 2024. Unifying sequences, structures, and descriptions for any-to-any protein generation with the large multimodal model HelixProTx. arXiv:2407.09274. Available at https://doi.org/10.48550/arXiv.2407.09274 [posted 12 July 2024; accessed 11 April 2025].

Diab, M., J. Herrera, B. Chernow, and C. Mao. 2022. Stable diffusion prompt book. Available at https://openart.ai/promptbook [accessed 11 April 2025].

García-González, A., J. Velasco, L. Velasco, and M. Ruiz-Mendez. 2018. An analytical simplification for faster determination of fatty acid composition and phytosterols in seed oils. *Food Analytical Methods* 11: 1234–1242.

González-Márquez, R., L. Schmidt, B. Schmidt, P. Berens, and D. Kobak. 2024. The landscape of biomedical research. *Patterns* 5(6): 100968.

Goodstein, D., S. Shu, R. Howson, R. Neupane, R. D. Hayes, J. Fazo, T. Mitros, et al. 2012. Phytozome: A comparative platform for green plant genomics. *Nucleic Acids Research* 40: D1178–D1186.

Grabb, D. 2023. The impact of prompt engineering in large language model performance: A psychiatric example. *Journal of Medical Artificial Intelligence* 6: 20. https://doi.org/10.21037/jmai-23-71

Guo, Z., K. Guo, B. Nan, Y. Tian, R. Iyer, Y. Ma, O. Wiest, et al. 2023. Graph-based molecular representation learning. arXiv:2207.04869. Available at https://doi.org/10.48550/arXiv.2207.04869 [posted 8 July 2022; accessed 11 April 2025].

Hassanein, M., A. Abdel-Razek, M. Rudzinska, A. Siger, K. Ratusz, and R. Przybylski. 2014. Phytochemical contents and oxidative stability of oils from non-traditional sources. *European Journal of Lipid Science and Technology* 116: 1563–1571.

Hawkins, C., D. Ginzburg, K. Zhao, W. Dwyer, B. Xue, A. Xu, S. Rice, et al. 2021. Plant metabolic network 15: A resource of genome-wide metabolism databases for 126 plants and algae. *Journal of Integrative Plant Biology* 63: 1888–1905.

Herck, J. V., M. V. Gil, K. M. Jablonka, A. Abdran, A. S. Anker, M. Asgari, B. Blaizik, et al. 2025. Assessment of fine-tuned large language models for real-world chemistry and material science applications. *Chemical Science* 16: 670–684. https://doi.org/10.1039/D4SC04401K

Horai, H., M. Arita, S. Kanaya, Y. Nihei, T. Ikeda, K. Suwa, Y. Ojima, et al. 2010. MassBank: A public repository for sharing mass spectral data for life sciences. *Journal of Mass Spectrometry* 45: 703–714.

Islam, M., B.-G. Jeong, J. Jung, E.-C. Shin, S.-G. Choi, and J. Chun. 2017. Phytosterol determination and method validation for selected nuts and seeds. *Food Analytical Methods* 10: 3225–3234.

Kamal-Eldin, A., and A. Moazzami. 2009. Plant sterols and stanols as cholesterol-lowering ingredients in functional foods. *Recent Patents on Food, Nutrition & Agriculture* 1: 1–14.

Kanehisa, M., S. Goto, M. Furumichi, M. Tanabe, and M. Hirakawa. 2012. KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research* 40: D109–D114.

Kassambara, A. 2024. statix: Pipe-friendly framework for basic statistical tests. R package version 0.7.2. Available at https://rpkgs.datanovia.com/rstatix/ [accessed 11 April 2025].

Kozlowska, M., E. Gruczynska, I. Scibisz, and M. Rudzinska. 2016. Fatty acids and sterols composition, and antioxidant activity of oils extracted from plant seeds. *Food Chemistry* 213: 450–456.

Lewis, P., E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, et al. 2021. Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Information Processing Systems 33 (NeurIPS 2020). Curran Associates, Newry, United Kingdom.

Li, H., J. Leung, and Z. Shen. 2024. Towards goal-oriented prompt engineering for large language models: A survey. arXiv:2401.14043. Available at https://doi.org/10.48550/arXiv.2401.14043 [posted 25 January 2024; accessed 11 April 2025].

Lim, S., K. Fo, R. Sunil, M. Itharajula, Y. Chuah, H. Foo, E. Davey, et al. 2023. PlantConnectome: Knowledge graph encompassing 70,000 plant articles. bioRxiv 2023.07.11.548541. Available at https://doi.org/10.1101/2023.07.11.548541 [posted 26 September 2024; accessed 11 April 2025].

Luo, H., P. Liu, and S. Esping. 2023. Exploring small language models with prompt-learning paradigm for efficient domain-specific text classification. arXiv:2309.14779. Available at https://doi.org/10.48550/arXiv.2309.14779 [posted 26 September 2023; accessed 11 April 2025].

Ngila, J., S. Yeboah, L. Wessjohann, and J. Schmidt. 2009. Profiling of phytosterols, tocopherols and tocotrienols in selected seed oils from Botswana by GC-MS and HPLC. *Journal of the American Oil Chemists' Society* 86: 617–625.

Nguyen, T., L. Doan, T. Thi, H. Tran, Q. Pham, H. Pham, L. Bach, et al. 2020. Fatty acids, tocopherols, and phytosterol composition of seed oil and phenolic compounds and antioxidant activity of fresh seeds from three *Dalbergia* species grown in Vietnam. *Processes* 8: 542.

Park, G., B.-J. Yoo, X. Luo, V. López-Marrero, S. Yoo, and S. Jha. 2023. Comparative performance evaluation of large language models for extracting molecular interactions and pathway knowledge. arXiv:2307.08813. Available at https://doi.org/10.48550/arXiv.2307.08813 [posted 17 July 2023; accessed 14 April 2025].

Qian, H., and Y. Jin. 2015. An updated megaphylogeny of plants, a tool for generating plant phylogenies and an analysis of phylogenetic community structure. *Journal of Plant Ecology* 9: 233–239. https://doi.org/10.1093/jpertv047

Rutz, A., M. Sorokina, J. Galgonek, D. Mietchen, E. Willighagen, A. Gaudry, J. G. Graham, et al. 2022. The LOTUS initiative for open knowledge management in natural products research. *eLife* 11: e70780. https://doi.org/10.7554/eLife.70780

Ryan, E., K. Galvin, T. O'Connor, A. Maguire, and N. O'Brien. 2007. Phytosterol, squalene, tocopherol content and fatty acid profile of

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selected seeds, grains, and legumes. *Plant Foods for Human Nutrition* 62: 85–91.

Saini, R., M.-H. Song, J.-W. Yu, X. Shang, and Y.-S. Keum. 2021. Phytosterol profiling of Apiaceae family seeds spices using GC-MS. *Foods* 10: 2378.

Samusevich, R., T. Hebra, R. Bushuiev, A. Bushuiev, T. Čalounova, H. Smrčkova, R. Chatpatanasiri, et al. 2024. Highly accurate discovery of terpene synthases powered by machine learning reveals functional terpene cyclization in Archaea. biorXiv 2024.01.29.577750. Available at https://doi.org/10.1101/2024.01.29.577750 [posted 25 April 2024; accessed 14 April 2025].

Schick, T., and H. Schütze. 2021. It's not just size that matters: Small language models are also few-shot learners. arXiv 2009.07118. Available at https://doi.org/10.48550/arXiv.2009.07118 [posted 15 September 2020; accessed 14 April 2025].

Schulhoff, S., M. Ilie, N. Balepur, K. Kahadze, A. Liu, C. Si, Y. Li, et al. 2024. The Prompt Report: A systematic survey of prompting techniques. arXiv 2406.06608v2. Available at https://arxiv.org/html/2406.06608v1 [accessed 14 April 2025].

Silva, S., G. Sampaio, and E. Torres. 2020. Phytosterols content in vegetable oils of Brazil: Coconut, safflower, linseed and evening primrose. *Brazilian Archives of Biology and Technology* 63: e20190216.

Smith, N., X. Yuan, C. Melissinos, and G. Moghe. 2024. Funcfetch: An LLM-assisted workflow enables mining thousands of enzyme-substrate interactions from published manuscripts. *Bioinformatics* 41: btae756.

Stuchlik, M., and S. Zak. 2002. Vegetable lipids as components of functional foods. *Biomedical Papers* 146: 3–10.

UniProt Consortium. 2015. UniProt: A hub for protein information. *Nucleic Acids Research* 43: D204–D212.

Wang, M., J. Carver, V. Phelan, L. Sanchez, N. Garg, Y. Peng, D. Nguyen, et al. 2016. Sharing and community curation of mass spectrometry data with Global Natural Products Social Molecular Networking. *Nature Biotechnology* 34: 828–837.

Wang, K., J. Li, N. Bhatt, Y. Xi, Q. Liu, U. Topcu, and Z. Wang. 2024. On the planning abilities of OpenAI's O1 models: Feasibility, optimality, and generalizability. arXiv 2409.19924. Available at https://doi.org/10.48550/arXiv.2409.19924 [posted 30 September 2024; accessed 14 April 2025].

Wei, J., X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *In* Advances in Neural Information Processing Systems 35 (NeurIPS 2022). Curran Associates, Newry, United Kingdom.

Wickham, H. 2016. ggplot2: Elegant graphics for data analysis. Springer-Verlag, New York, New York, USA.

Wickham, H., M. Averick, J. Bryan, W. Chang, L. McGowan, R. Francois, G. Grolemund, et al. 2019. Welcome to the tidyverse. *Journal of Open Source Software* 4(43): 1686.

Winter, D. 2017. rentrez: An R package for the NCBI EUtils API. *The R Journal* 9: 520–526.

Wu, S., Y. Xiong, Y. Cui, H. Wu, C. Chen, Y. Yuan, L. Huang, et al. 2024. Retrieval-augmented generation for natural language processing: A survey. arXiv:2407.13193. Available at https://doi.org/10.48550/arXiv.2407.13193 [posted 18 July 2024; accessed 14 April 2025].

Xu, M., X. Yuan, S. Miret, and J. Tang. 2023. ProtST: Multi-modality learning of protein sequences and biomedical texts. *Proceedings of the 40th International Conference on Machine Learning* 202: 38749–38767.

Yang, R., L. Xue, L. Zhang, X. Wang, X. Qi, J. Jiang, L. Yu, et al. 2019. Phytosterol contents of edible oils and their contributions to estimated phytosterol intake in the Chinese diet. *Foods* 8: 334.

Yang, X., J. Gao, W. Xue, and E. Alexandersson. 2024. PLLama: An open-source large language model for plant science. arXiv:2401.01600. Available at https://doi.org/10.48550/arXiv.2401.01600 [posted 3 January 2024; accessed 14 April 2025].

Yao, S., D. Yu, J. Zhao, I. Shafran, T. Griffiths, Y. Cao, and K. Narasimhan. 2023. Tree of Thoughts: Deliberate problem solving with large language models. *In* Advances in Neural Information Processing Systems 36 (NeurIPS 2023). Curran Associates, Newry, United Kingdom.

Yu, G., D. Smith, H. Zhu, Y. Guan, and T. Lam. 2017. ggtree: An R package for visualization and annotation of phylogenetic trees with their covariates and other associated data. *Methods in Ecology and Evolution* 8: 28–36.

Zhang, L., T. Ergen, L. Logeswaran, M. Lee, and D. Jurgens. 2024. Sprig: Improving large language model performance by system prompt optimization. arXiv:2410.14826. Available at https://doi.org/10.48550/arXiv.2410.14826 [posted 18 October 2024; accessed 14 April 2025].

Zhong, Z., K. Zhou, and D. Mottin. 2024. Benchmarking large language models for molecule prediction tasks. arXiv:2403.05075. Available at https://doi.org/10.48550/arXiv.2403.05075 [posted 8 March 2024; accessed 14 April 2025].

Zhou, Y., A. Muresanu, Z. Han, K. Paster, S. Pitis, H. Chan, and J. Ba. 2023. Large language models are human-level prompt engineers. OpenReview.net. Available at https://openreview.net/forum?id=92gvk82DE- [accessed 14 April 2025].

## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

**Appendix S1.** Prompts used in identifying enzyme–product pairs.

**Appendix S2.** Prompts used in identifying compound–species pairs.

**Appendix S3.** Comparison of language model evaluation of compound–species associations versus co-occurrence in text.

**Appendix S4.** Comparison of PyTesseract processing of table images and multimodal language model processing of table images.

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